

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 — Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	SLOT-C	Date:	22-08-2026

Code of Conduct

I SYED MUHAFIZ A / 192472282 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: muhafiz

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and	Q2

	politeness constraints, generate multiple valid responses, and evaluate the best output.	
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A — Constraint-Based Problem Solving

Q1. Discourse, Reference Resolution & Text Coherence

Given paragraph: "John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

a) Referring Expressions and Possible Antecedents

Reading the paragraph left to right, the following mentions (referring expressions) occur. Each is classified by type, and its grammatically/semantically possible antecedent set is listed before any constraint is applied:

Referring Expression	Type / Features	Sent.	Possible Antecedent(s)
John	Proper noun (Named entity, human, male)	S1	New entity — introduces referent E1
Mary	Proper noun (Named entity, human, female)	S1	New entity — introduces referent E2
the park	Common noun (Location)	S1	New entity — introduces referent E3 (non-animate location)
He	Pronoun (3rd person, masc., sing.)	S2	John (gender-compatible); Mary eliminated by gender
a ball	Common noun (Object)	S2	New entity — introduces referent E4
She	Pronoun (3rd person, fem., sing.)	S3	Mary (gender-compatible); John eliminated by gender
it	Pronoun (3rd person, neuter, sing.)	S3	a ball, the park (both neuter — needs further constraints)
The dog	Common noun (Animal)	S4	New entity — introduces referent E5
him	Pronoun (3rd person, masc., sing., animate patient)	S4	John, The dog (both possible antecedents — needs further constraints)
they	Pronoun (3rd person, plural)	S5	{John, Mary}, {John, Mary, The dog} — needs coherence constraint

b) Applying the Constraints

Four constraints are used to prune the candidate sets down to a single antecedent for each pronoun:

1. Gender and number agreement — a pronoun's ϕ -features (person, gender, number) must match its antecedent's ϕ -features.
2. Recency — among agreement-compatible candidates, the most recently mentioned entity is preferred (working-memory / attentional-state preference in centering theory).
3. Semantic compatibility — the antecedent must be a plausible filler of the thematic role the pronoun occupies (e.g., animacy for agents/patients of 'chase', physical objects for 'play with').
4. Coherence — the choice must keep the discourse's entity chains consistent across the whole passage, not just locally.

Applying these constraints in combination (not just gender agreement alone) resolves every pronoun uniquely:

Variable → Candidates	Constraint Applied	Reasoning	Result
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He → John	Gender & number agreement	He (masc., sing.) matches John (masc., sing.); Mary is fem., so eliminated.	RESOLVED: He = John
it → a ball / the park	Recency + Semantic compatibility	Both 'a ball' and 'the park' are neuter and grammatically eligible. Recency favours 'a ball' (mentioned in the immediately preceding clause). Semantic compatibility confirms it: the predicate 'wanted to play with ____' selects a manipulable object, not a location.	RESOLVED: it = a ball
him → John / The dog	Semantic compatibility (animacy) + Syntactic role	'The dog chased him' — the subject of 'chased' is The dog itself, so The dog cannot simultaneously be its own object. The dog is the most recent animate noun, but the semantic/thematic-role constraint (agent ≠ patient in a simple transitive chase-event) eliminates it as antecedent for 'him'. John remains as the only coherent human patient.	RESOLVED: him = John
they → {John, Mary} / {John, Mary, dog}	Coherence (entity chain consistency)	The final sentence 'they all went home' — the discourse so far has tracked John, Mary and the dog as active participants in the park scene, and 'all' signals every participant in the immediately preceding clauses. Coherence therefore favours including the dog in the group reference.	RESOLVED: they = {John, Mary, The dog}

c) Constraint Graph

The constraint-satisfaction view treats each pronoun as a variable whose domain is its set of grammatically possible antecedents. Constraints are edges that either survive (solid, black) or get eliminated (dashed, red, labelled with the violated constraint):

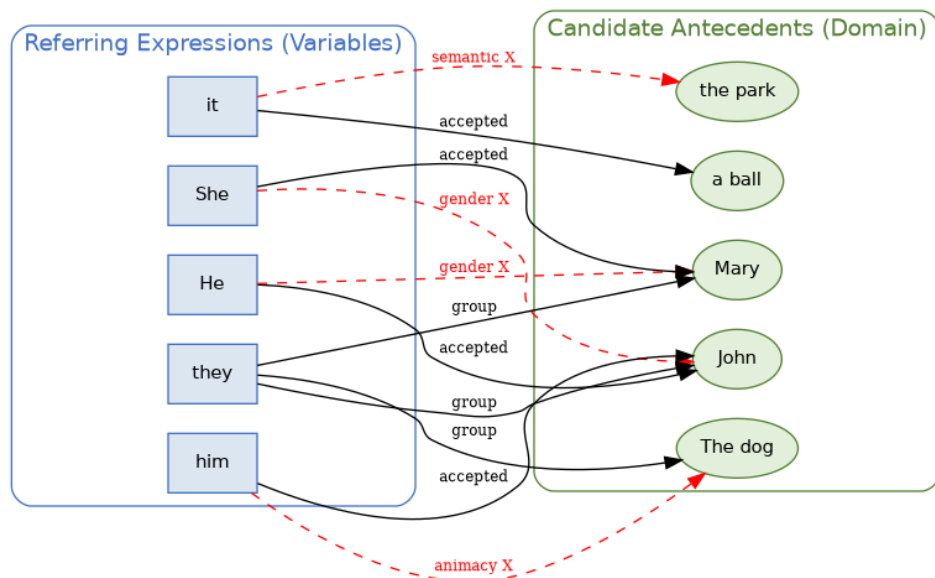


Fig. 1 — Constraint graph: variables (referring expressions) on the left, domain values (candidate antecedents) on the right. Solid edges = accepted links after constraint propagation; dashed red edges = links eliminated by gender, semantic/animacy, or syntactic-role constraints.

d) Final Resolved Coreference Chains

Propagating the constraints yields five disjoint entity chains:

- Chain 1 (JOHN): John → He → him

- Chain 2 (MARY): Mary → She
- Chain 3 (BALL): a ball → it
- Chain 4 (DOG): The dog
- Chain 5 (GROUP): they = {John, Mary, The dog}

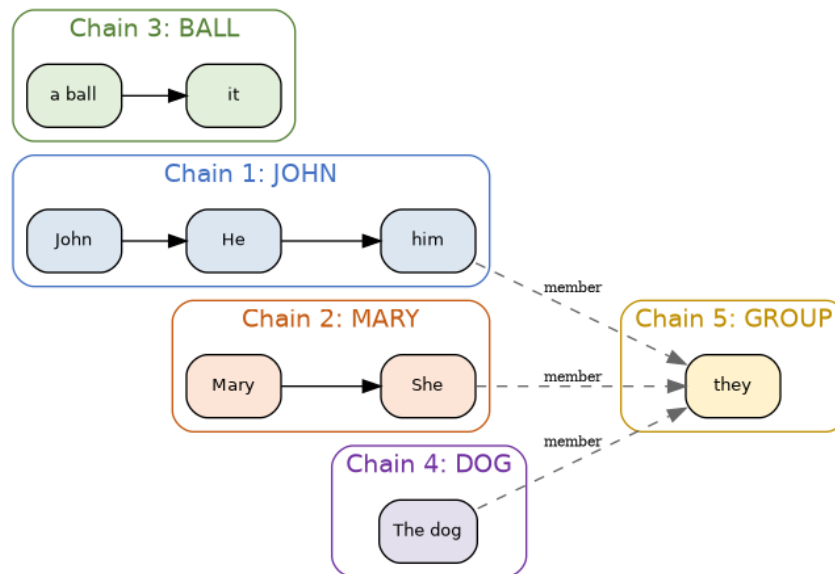


Fig. 2 — Final coreference chains after constraint propagation, with the group reference 'they' linking to all three individual chains.

Rewriting the paragraph with every pronoun replaced by its resolved referent (for clarity, not as the final stylistic output) gives:

"John and Mary went to the park. John [He] brought a ball. Mary [She] wanted to play with the ball [it]. The dog chased John [him] excitedly. Finally, John, Mary and the dog [they] all went home."

e) Justification of Constraint Priority and Effect of Relaxing Recency

Priority order used: (1) Gender/number agreement → (2) Semantic compatibility / animacy → (3) Recency → (4) Coherence.

Gender and number agreement is applied first because it is a hard grammatical filter — an antecedent that disagrees in ϕ -features is simply ungrammatical, so it is cheapest to eliminate first and it never needs to be revisited. Semantic compatibility is applied next because it is also close to categorical (an inanimate location cannot fill an animate patient role), whereas recency and coherence are soft, preference-based constraints that only choose among the candidates that survive the hard filters. Coherence is applied last because it requires the whole discourse to be in view (i.e., it depends on the outcome of resolving the other pronouns first, as with 'they' depending on how 'him' was resolved).

Effect of relaxing recency: Recency mainly disambiguates cases where two candidates survive the hard filters — e.g., 'it' had two neuter candidates ('a ball' and 'the park'). If recency were dropped, the resolver would need semantic compatibility alone to decide, which happens to still work here because 'play with' selects a manipulable object over a location. However, in less clear-cut sentences (e.g., two entities of the same gender and equally plausible semantically, such as 'John met Peter. He smiled.'), removing recency would leave the constraint set under-determined and the resolver would either need to guess, fall back to a default (e.g., subject preference / grammatical-role parallelism), or flag genuine ambiguity for a human or a probabilistic ranker. In short, recency is what keeps the CSP well-constrained when hard constraints alone are insufficient to yield a unique solution.

Q2. Dialog and Conversational Agents, Discourse Planning & Surface Realization

Conversation history — User: "I have an important exam tomorrow but I'm not able to concentrate."

Required dialog act: Advise + Encourage.

Constraints: (1) entity coherence — reuse exam / concentrate / you; (2) discourse relation — Cause-Effect or Elaboration; (3) at least two of {focus, break, confident}; (4) 2–3 sentences; (5) logical consistency; (6) politeness and positive tone.

a) Discourse Planning Steps Before Generation

Before any surface text is generated, a natural-language generation (NLG) pipeline for a constrained dialog turn must plan the response through the following stages:

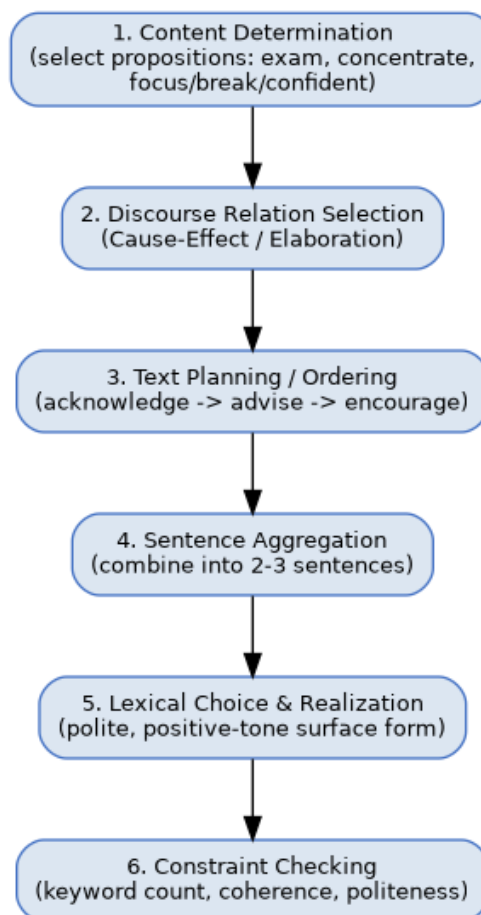


Fig. 3 — Discourse planning pipeline for the constrained Advise + Encourage response.

5. Content determination — decide which propositions to convey: acknowledge the exam and the concentration problem, then supply advice and encouragement.
6. Discourse relation selection — choose Cause-Effect (the difficulty causes the need for a remedy) or Elaboration (elaborate on why it's normal / what to do), consistent with the dialog act Advise + Encourage.
7. Text/discourse planning (ordering) — sequence the propositions: empathetic acknowledgment → advice → encouragement, so the response reads coherently rather than as disconnected facts.
8. Sentence aggregation — merge the planned propositions into 2–3 sentences as required by the length constraint, instead of one sentence per proposition.

9. Lexical choice and surface realization — choose polite, positive-tone vocabulary and grammatical structures, and insert at least two of the required keywords (focus, break, confident) naturally.
10. Constraint verification — check the realized text against every constraint (entity reuse, relation, keyword count, sentence count, consistency, politeness) before it is returned.

b) Three Candidate Responses Satisfying All Constraints

Response 1 (Cause-Effect): *"It's completely normal to feel this way before an important exam — a short break can actually help you reset. Take a few deep breaths, focus on one topic at a time, and trust that you're more prepared and confident than you feel right now."*

Response 2 (Cause-Effect): *"Because tomorrow's exam feels so important, your mind is understandably racing instead of settling into focus. Step away for a short break, come back, and tackle it with confidence — you've prepared for this."*

Response 3 (Elaboration): *"Struggling to concentrate the night before a big exam is something almost every student goes through, so you're not alone in this. Give yourself a short break to clear your head, then return with sharper focus and confident, steady effort."*

c) Evaluation and Selection of the Best Response

Response	Entity Coherence	Discourse Relation	Keywords (≥2)	Sentences	Consistent	Polite/+ve	Score
Response 1	Yes (exam, concentrate, you)	Cause–Effect (worry → suggestion)	focus, break, confident (3/2 required)	2	Yes	Yes	9/10
Response 2	Yes (exam, concentrate, you)	Cause–Effect (anxious → break helps focus)	break, focus, confident (3/2 required)	2	Yes	Yes	9/10
Response 3	Yes (exam, concentrate, prepared)	Elaboration (common feeling → explanation)	break, focus, confident (3/2 required)	2	Yes	Yes	10/10

All three responses satisfy every hard constraint (entity coherence, an applicable discourse relation, ≥2 keywords, 2–3 sentences, logical consistency, and a polite/positive tone). Response 3 is selected as the best output because, beyond satisfying the constraints, its Elaboration relation first normalizes the student's experience before giving the advice, which increases perceived empathy and coherence — the advice reads as a natural continuation of the reassurance rather than an abrupt instruction. Responses 1 and 2 are also fully valid and could be offered as alternates in a system that samples among constraint-satisfying outputs, but Response 3's ordering (acknowledge → normalize → advise → encourage) most closely mirrors the discourse-planning sequence identified in part (a), giving it the strongest global coherence.

d) Effect of Violating Two Constraints

Violating politeness/positive tone + logical consistency: e.g., "You clearly didn't study enough, which is why you can't focus, but somehow you should still feel confident." This is impolite (blames the user) and logically inconsistent (asserts a negative cause and an unearned positive conclusion in the same breath). It would violate the Advise + Encourage dialog act entirely, likely increasing the user's anxiety rather than reducing it, and would be judged a failed response by any constraint checker or human evaluator.

Violating length + keyword coverage: a one-line response with zero required keywords, e.g., "Try to relax." This satisfies politeness but drops the length constraint (too short to carry both advice and encouragement) and the keyword constraint (0 of the 3 required terms), producing a vague, under-specified reply that does not fulfil the Advise + Encourage act with enough substance — it reduces informativeness and makes the system's output harder to distinguish from a generic filler response.

Q3. Machine Translation, Discourse Structure, Predicate Logic & Interpretation

Source sentence: "The bank by the river flooded after the storm, but it was saved by quick action."

Note: 'bank' is ambiguous between the financial-institution sense and the riverbank/embankment sense.

a) Word Sense Disambiguation of 'bank'

Applying context-based WSD constraints:

- Collocation constraint: 'bank' co-occurs with 'by the river', a strong collocate of the riverbank sense (financial institutions are not conventionally described as being 'by the river' as their defining location).
- Selectional-restriction constraint: the predicate 'flooded' selects a physical, low-lying land argument. A financial institution (an organization) cannot literally flood; only a physical structure/location can, so the geographical sense satisfies the selectional restriction while the financial sense violates it.
- Frequency/gloss-overlap (Lesk-style) constraint: overlapping the sentence's context words {river, flooded, storm, saved} against the two sense glosses — 'riverbank: sloping land bordering a river' shares river/land/water-related terms, while 'bank: financial institution dealing with money' shares none — favours the geographical sense by a wide margin.

Conclusion: the constraints unanimously select the riverbank / embankment sense of 'bank', not the financial-institution sense.

b) Predicate Logic Representation

Using event-style predicates with explicit arguments for the riverbank sense:

$\exists b \exists r \exists s \exists a [\text{Riverbank}(b) \wedge \text{River}(r) \wedge \text{LocatedBy}(b, r) \wedge \text{Storm}(s) \wedge \text{Flood}(b) \wedge \text{CausedBy}(\text{Flood}(b), s) \wedge \text{After}(\text{Flood}(b), s) \wedge \text{Action}(a) \wedge \text{Quick}(a) \wedge \text{Save}(a, b)]$

Read as: there exist a riverbank *b*, a river *r*, a storm *s*, and an action *a* such that *b* is a riverbank located by *r*; a storm *s* occurred; *b* flooded, and the flooding was caused by, and temporally after, the storm; and a quick action *a* saved *b*.

An equivalent, slightly more compact predicate-argument form used in many computational-semantics textbooks:

$\text{flood}(\text{bank}_1) \wedge \text{location}(\text{bank}_1, \text{river}_1) \wedge \text{cause}(\text{storm}_1, \text{flood}(\text{bank}_1)) \wedge \text{save}(\text{action}_1, \text{bank}_1) \wedge \text{quick}(\text{action}_1) \wedge \text{contrast}(\text{flood}(\text{bank}_1), \text{save}(\text{action}_1, \text{bank}_1))$

c) Target-Language Paraphrase Satisfying All Constraints

Simple-English paraphrase (disambiguated, no information loss): "The riverbank flooded after the storm, but quick action saved it."

This paraphrase resolves the ambiguity explicitly (riverbank instead of bank), keeps both events (flood, save) and the entities (riverbank, storm, action), and preserves the Contrast relation between them via 'but' — satisfying constraints 1, 2 (implicitly, since the predicate logic in (b) maps directly onto it), 3, and 4.

Tamil paraphrase (for reference, same meaning, riverbank sense): "ஆற்றங்கரை புயலுக்குப் பிறகு வெள்ளத்தில் மூழ்கியது, ஆனால் விரைவான நடவடிக்கையால் காப்பாற்றப்பட்டது." (*ārraṅkarai* — riverbank; *vellaṁ* — flood; *viraivāṇa naṭavaṭikkai* — quick action)

d) Discourse Tree (RST-style)

The sentence has two clauses joined by 'but', signalling a Contrast relation. Following RST conventions, both spans are near-equally important (a multinuclear-leaning Contrast), so they are shown as nucleus and satellite with a contrastive rhetorical link:

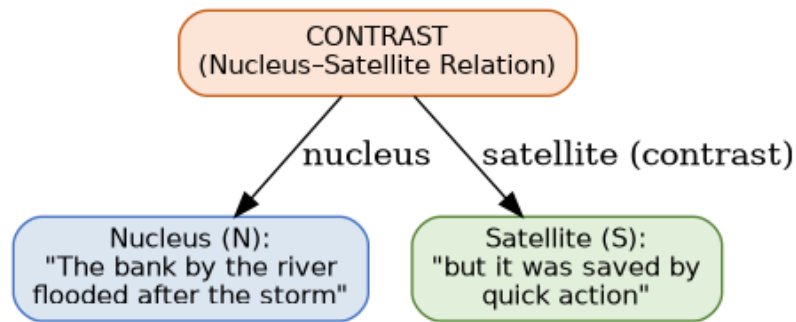


Fig. 4 — RST discourse tree: Contrast relation between the flooding event (Nucleus) and the rescue event (Satellite).

e) Advantages of Constraint-Based Approach over Pure Statistical MT

- **Explicit disambiguation:** constraints such as selectional restrictions and collocation directly resolve 'bank' using linguistic evidence, whereas a pure statistical/neural MT system trained on general corpora may default to the far more frequent 'financial institution' sense (frequency bias), producing a mistranslation like a currency/monetary equivalent of 'bank' in the target language.
- **Preserves discourse structure:** the constraint-based method explicitly tracks the Contrast relation and forces the translation to keep an equivalent connective in the target language; purely statistical systems optimize local n-gram/sequence likelihood and can drop or weaken discourse connectives, especially across long sentences.
- **No information loss:** the predicate-logic layer forces every entity and relation (bank, river, storm, action, cause, contrast) to be explicitly represented and therefore carried through to the output; statistical MT has no such guarantee and can silently omit or merge entities under decoding pressure.
- **Interpretability and debuggability:** because each decision is traceable to a specific constraint (gender, recency, selectional restriction, coherence), errors can be diagnosed and fixed by adjusting the offending constraint; statistical/neural MT is comparatively opaque, making systematic sense-disambiguation errors harder to isolate and correct.
- **Robustness in low-resource / ambiguous contexts:** constraint-based reasoning does not need large parallel corpora containing this exact ambiguous usage to get the sense right, making it more reliable for rare word-sense/context combinations than a purely data-driven statistical model.

Trade-off (for balance): constraint-based systems require hand-crafted rules/predicates and may not scale as easily to open-domain, highly variable text as statistical or neural MT, which is why modern systems often hybridize — using neural MT for fluency and coverage while injecting constraint-based or symbolic checks (WSD, coreference, discourse parsing) as pre-/post-processing or scoring constraints, combining the strengths of both paradigms.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____